Emergency Response App — Cardiovascular Hospital

Product Overview

A mobile emergency response system used by hospital staff to receive cardiac alerts, understand patient context, and coordinate response during high-stress scenarios. The hospital already had an existing emergency app built by another vendor. The objective was not to rebuild it, but to simplify it for real emergency behavior, especially during night shifts.

My Role

UX/UI Designer collaborating with Product, Engineering, nurse heads, and ward coordinators. Focused on restructuring alert and communication flows, redesigning critical emergency surfaces, improving information hierarchy, and iterating through prototype walkthroughs with hospital staff.

The Real Problem

In cardiac emergencies, the first 30 seconds determine response quality. In the existing system, alerts displayed vitals, history, chat, and actions simultaneously. Nurses had to interpret data while deciding what action to take. Communication and alerts were tightly mixed. Dense layouts slowed recognition, especially during night shifts. Nothing was technically broken, but cognitive load was highest exactly when clarity mattered most.

Core Design Principle

Separate scanning from acting. Users should first understand what is happening, then decide, and then act — not perform all three simultaneously.

Structural Shift: Alert → Scan → Act

The experience was restructured into two clear layers. The Scan Layer presents a large alert card containing patient vitals, location, alert severity, and elapsed time — without actions or secondary information. Only after context is understood does the Action Layer provide access to calling, team chat, and detailed patient history. This reduced hesitation and prevented premature interaction during high-stress moments.

Information Architecture (Insert Diagram)

Interface Design — High-Fidelity Screens (Insert Annotated Screens)

Readability Under Pressure

Stakeholders emphasized the need for readability during night shifts. Initial iterations increased font size slightly, which proved insufficient. The final approach increased overall card scale, reduced information density, and embraced vertical space since nurses handle one case at a time. Improvement came from layout hierarchy rather than visual embellishment.

Constraints

Workflow ambiguity persisted for 1–2 months due to evolving AI discussions. Mid-project, the technology stack shifted from React + Material to Ionic, requiring alignment with new component patterns. Instead of redesigning the full visual system, focus was placed on surfaces that directly influenced emergency response behavior.

Validation

The redesigned flow went through three prototype iterations involving five nurses and two ward coordinators. Early versions failed due to insufficient visual scale. Later iterations aligned around speed of recognition and reduction of cognitive effort.

Outcome

The product shipped and went live within hospital workflows. The alert flow consistently met the sub-30-second response target. Staff described the experience as more comfortable than existing complex systems during emergency contexts. The impact was behavioral — reduced hesitation and faster clarity under pressure.